Original Query (Q)

```
1: SELECT COUNT(*)
2: FROM PostHistory ph,
3:   Posts p,
4:   Users u,
5:   Badges b
6: WHERE b.UserId = u.Id
7:   AND ph.OwnerUserId = u.Id
8:   AND ph.UserId = u.Id
9:   AND ph.CreationDate >= '2010-07-27 18:08:19'
10:  AND ph.CreationDate <= '2014-09-10 08:22:43'
11:  AND p.PostTypeId = 2;
```

ROLEX Engine

Fragment Index (FI)

$\mathcal{K}_1$: AGG(JOIN[INNER])
$\mathcal{K}_2$: JOIN[INNER](JOIN,SCAN)
$\mathcal{K}_3$: JOIN[INNER](SCAN,SCAN)

FI lookup

R_1: JoinReorder
R_2: PredicateDeriveTransitive
R_3: JoinDecomposeToPreaggregatedCount

MCTS Policy $\Rightarrow \{R3 \rightarrow R2 \rightarrow R1\}$

Rewritten Query (Q')

```
1: WITH ph_counts AS (
2:   SELECT UserId, COUNT(*) AS ct
3:   FROM PostHistory
4:   WHERE CreationDate BETWEEN '2010-07-27 18:08:19' AND '2014-09-10 08:22:43'
5:   GROUP BY UserId),
6: p_counts AS (
7:   SELECT OwnerUserId, COUNT(*) AS ct
8:   FROM Posts WHERE PostTypeId = 2 GROUP BY OwnerUserId),
9: b_counts AS (SELECT UserId, COUNT(*) AS ct FROM Badges GROUP BY UserId)
10: SELECT SUM(ph.ct * p.ct * b.ct)
11: FROM ph_counts AS ph
12:   INNER JOIN p_counts  AS p ON ph.UserId = p.OwnerUserId
13:   INNER JOIN b_counts  AS b ON ph.UserId = b.UserId;
```